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Abstract

Drawing on the professions and institutional literature, we develop theory about how professional work can reflect multiple institutional logics by analyzing changes in the work of pharmacists over time. Through a historical case study of U.S. pharmacists from 1852 to the present, we propose a new conceptualization of professionals and professional work as guided by a constellation of logics derived from broader society. We show that both competitive (segmenting) and cooperative (facilitative or additive) relationships among coexisting logics allow for the simultaneous influence of multiple logics on professionals and their work.

Keywords

professionals, professional work, institutional logics

In recent years, there has been a profound transformation in the landscape of professional work. Professional firms have consolidated, growing larger and more global in scope. New organizational forms have arisen that combine professional groups, which traditionally have had strict jurisdictional boundaries.

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Rather than being valued primarily for their expertise, professionals are increasingly expected to meet standards of both effectiveness and efficiency. And increasingly professionals work as employees of large organizations where their status as independent professionals is less clear.

Perhaps one of the most extreme cases of transformation in professional work is visible in the case of pharmacists. As a result of technological advancements, their work changed dramatically from the mid-1800s, when they concocted and sold relatively untested remedies, to today, when pharmacists dispense prepackaged, mass produced, quality-assured medications. U.S. pharmacists formerly worked as independent shopkeepers until the rise of large drugstores and mass merchandisers such as Walmart led to the current situation where only 3% of retail pharmacists are independent store owners (U.S. Department of Health & Human Services, 2000). Although pharmacists may have experienced more changes than other professionals, their experiences are similar to others faced with increasing market pressures, changing work, and shifts to employee rather than independent owner status.

Sociologists have long been interested in professionals and professional work. They have developed the theory of professions in a way that is sensitive to providing a dynamic portrayal of professional development (Timmermans, 2008). In discussing the status of professions in advanced industrial societies, many scholars have theorized about professionalism and the inherent threats to its privileged position from a variety of social and economic changes (e.g., Freidson, 1993; Krause, 1996; Leicht & Fennel, 1997; Timmermans, 2008). While several scholars have formalized these challenges to professionalism in terms of different logics (e.g., Freidson, 2001; Light, 1989), there has been no explanation of how multiple competitive logics can coexist and simultaneously influence professional work.

The concept of logics has become central to institutional theory. Logics were introduced into institutional theory by Friedland and Alford (1991) as part of their attempt to emphasize the importance of social context. They argued that each societal sector is characterized by a central logic that is associated with "a set of material practices and symbolic constructions," which shape and constrain the behavioral repertoire of social actors (1991, p. 248). Although most institutional scholars recognize that multiple logics coexist (e.g., Scott, 2008a; Thornton & Ocasio, 2008), they have so far given little direct attention to ongoing situations where professionals (or any other single actor) are affected by coexisting multiple logics.

In this article, we build on and extend the professions and institutional literature to better understand the relationship of multiple logics with the changing nature of professional work. Our goal is to develop theory about

how different logics can collectively be reflected in professional work by examining a historical case study of U.S. pharmacists. Based on our analyses, we develop a new conceptualization of professionals and professional work as guided by a constellation of logics derived from broader society. We propose that this conceptualization provides a theoretical foundation to understand coexisting multiple logics and their simultaneous influence on professionals and their work.

Literature Review

Sociologists hold a long history of studying professionals and professional work. Some scholars have examined the role of professionals in society and attempted to identify patterns in the transition from occupational groups to professional status (e.g., Caplow, 1954; Carr-Saunders & Wilson, 1933; Wilensky, 1964). Others have approached professionalism as a form of social control, arguing that occupations have constantly pushed to become professionals as a way to gain a monopolistic position in the market place (e.g., Freidson, 1970; Larson, 1977). Yet other sociologists have studied the influence of gender, ethnic, and racial differences in obtaining professional status (e.g., Levin, 2001; Seron & Ferris, 1995; Wilson, Sakura-Lemessy, & West, 1999). What all these studies have in common, is an interest in understanding the process of professionalization.

However, even after an occupational group becomes professionalized, it continues to be subjected to external societal pressures. The importance of this ongoing process has been recognized in the theory of professions through attention to different ways in which professional development occurred. For example, scholars have examined how boundaries between occupational groups were negotiated (e.g., Abbott, 1988; Montgomery & Oliver, 2007; Nancarrow & Borthwick, 2005). They have also theorized how shifting social control at a societal level affected the privileged position of professionals (e.g., Light, 1993, 1995, 2000). Other scholars have focused on changes in professional work including actual content, financial reimbursement, and its organizational structure (Leicht & Fennel, 2008). They have examined changes in organizational settings where professionals work and discussed how control of professional work has become more within the realm of corporate and bureaucratic control (e.g., Freidson, 1993; Krause, 1996; Leicht & Fennel, 1997). They have also studied how the special economic and social position of professionals was reduced, and attributed this to the infringement of state and private capital on professional expertise in the name of competition and efficiency (Freidson, 2001; Krause, 1996). Finally, studies have

addressed how external pressures for accountability and cost containment resulted in direct incursions on professional control over core technology (Hafferty & Light, 1995; Leicht & Fennel, 1997, 2008).

Freidson (2001) brought many of these perspectives into one framework by proposing that professional work can be organized by three different logics: market, rational-legal bureaucracy, and professional. While grounded in the world of work, he abstracts these logics from reality to represent the different principles by which work could be organized. Freidson contrasts the attributes of *professionalism* with those of *bureaucracies*, based on Weber's (1949) model of rational-legal bureaucracy, and the *market*, based on Adam Smith's (1976) model of the free market. According to Freidson, members of an occupation control their own work in professionalism, the production of goods and services are planned and controlled by the administration of large organizations in rational-legal bureaucracies, and free and unregulated competition exists in markets. In explicating these differences between the logics, Freidson suggests that professional work today is organized by a combination of all three logics, but presents a model of competition between logics without developing theoretical arguments to explain their coexistence or mechanisms for managing their competitive aspects.

The question of multiple logics and how tensions between them are resolved has received more attention in the institutional literature. Friedland and Alford (1991) introduced the concept of logics to institutional theory as part of their concern that social context had been neglected. They argued that attention to societal level institutional orders was necessary to understand individual and organizational behavior, since each of the most important societal level institutional orders—market, state, democracy, family, and religion—is characterized by a central logic. While Freidson focused on how work is organized, institutional theorists have taken a broader perspective by drawing attention to the “central institutions of the contemporary capitalist West” (Friedland & Alford, 1991, p. 232) and by conceptualizing logics as hierarchical in form (Thornton & Ocasio, 2008). Organizational fields and industries are viewed as having their own logics nested within societal level institutional orders.

Similar to Freidson's conception of logics as guiding principles, logics in this tradition have been defined as “ways of ordering reality” (Friedland & Alford, 1991, p. 243), the “axial principles of organization and action” (Thornton, 2004, p. 2), and the “rules of action, interaction, and interpretation” (Thornton & Ocasio, 1999, p. 804). Logics shape individual and organizational practices because they represent sets of expectations for social relations and behavior. A core assumption is that the interests, identities, and

values of individuals and organizations are embedded in logics and provide the context for decisions and outcomes (Thornton & Ocasio, 2008). Logics provide social classification and categorization (DiMaggio, 1997), which have been linked to differences in work practices (Mohr & Duquenne, 1997), organizational forms (Haveman & Rao, 1997), and product categories (Lounsbury & Rao, 2004). Logics also shape practices by structuring individual and organizational attention vis-à-vis rules and conventions for deciding what issues are important and thus worth resolving (Thornton, 2002).

Although most discussions within an institutional approach recognize that multiple logics coexist (e.g., Scott, 2008a; Thornton & Ocasio, 2008), the focus has been on how a dominant logic organizes and guides the behavior of social actors (Greenwood, Diaz, Li, & Lorente, 2010; Lounsbury, 2007). Only recently have scholars begun to address the role of nondominant logics. So far, the literature presents three explanations concerning the relationships among multiple coexisting institutional logics: (a) Among existing logics, one is dominant and guides behavior, (b) multiple logics battle with each other for dominance, and (c) multiple logics differentially affect different actors, geographical communities, or organizations. We discuss each of these explanations below.

The first way of conceptualizing the relationship among multiple institutional logics is in terms of a dominant logic and secondary or subordinate logics that exist but do not guide behavior. Studies based on this conceptualization tend to investigate the consequences of shifts in the dominance of logics at a population level. They use subordinate (or secondary) logics to explain shifts from one dominant logic to another because they are viewed as the source of change. Based on underlying expectations of competition between logics, subordinate logics are presented as challengers that usurp previously dominant logics (e.g., Goodrick, 2002; Lounsbury, 2002; Thornton, 2002; Thornton & Ocasio, 1999). For example, Scott, Ruef, Mendel, and Caronna (2000) document the existence of both dominant and nondominant logics in each identified era of health system change.

A second explanation of the relationship among multiple institutional logics is that of long-term competition during a lengthy transition period where multiple competing logics exist and a resolution is required. In these studies, multiple logics are portrayed as battling with each other for supremacy until ultimately one logic wins out-becoming the new dominant institutional logic for a period of time (e.g., Hensman, 2003; Hoffman, 1999). The difference between this conceptualization and the previous one is the focus on a potentially long period of time of competing logics where different actors continue to hold different logics and where there is no winner. This could be a time of

battlefield mentality, or an uneasy truce (Reay & Hinings, 2005). More explicitly, this conceptualization suggests that a dominant or prevailing institutional logic always represents some type of truce or resolution between contradictory underlying logics (Suddaby & Greenwood, 2005). That is, competition or tension between multiple logics is continual.

Third, a few studies have suggested that coexisting and competing institutional logics do not always resolve. Instead, different logics may be associated with different actors, geographical communities, or types of organizations. Reay and Hinings (2009), for example, show how physicians and managers were each guided by different, competing logics over a lengthy period of time. Both Lounsbury (2007) and Greenwood et al. (2010) investigate how regional differences in logics means that organizational practices differ by region. Other scholars have shown how separate and competing logics held by different segments of an industry result in different practices (Marquis & Lounsbury, 2007; Thornton, Jones, & Kury, 2005). While all these studies provide evidence that different actors can be guided by different logics, the institutional literature has given little attention to how professionals (or any other single actor) are affected by an ongoing situation where multiple logics coexist.

In sum, neither the professional nor institutional literature has adequately addressed the relationship of multiple logics with the changing nature of professional work. While both literatures have recognized that multiple logics exist, neither literature has articulated how the work of a single group of professionals can reflect the influence of multiple logics. In the next section, we draw on both the professional and institutional literatures to develop a theoretical framework that allows us to better understand how different logics can collectively organize the work of a single profession.

Theoretical Framework

Previous scholars have identified logics and how they affect behavior. However, there has not been a systematic explication of the logics that are most likely to be reflected in professional work, and therefore we are missing a theoretical foundation for comparing the relative influence of these relevant logics. Institutionalists have explicated logics in terms of common attributes, which facilitates comparison. But their work has been focused at a societal level (e.g., nature of economic systems) and is not directly applicable to professional work (e.g., Thornton, 2004). Not surprisingly, scholars studying professionals have been more interested in applying logics to professional work but they have either focused on the unique attributes of different logics (e.g., Freidson, 2001) or engaged in a narrow comparison of two

logics in a limited domain (e.g., Light, 1989). Building on previous conceptions of logics in terms of ideal types (Freidson, 2001; Light, 1989; Rao, Monin, & Durand, 2003; Thornton, 2004; Thornton & Ocasio, 2008), we describe below the characteristics of four logics, which may be reflected in professional work: the professional, corporate, market, and state logics.

To identify the ideal type for each logic, we drew on our knowledge of the institutional and professions literature. We turned our attention to previous studies where ideal types were developed to study institutional logics (e.g., Freidson, 2001; Light, 1989; Rao et al., 2003; Thornton, 2004; Thornton & Ocasio, 2008). Ideal types are abstractions from reality and represent a "pure case in which the relevant features are distinct and unambiguous" Weber, 1949, p. 88). By providing a stable point, ideal types facilitate systematic comparison of empirical variation (Freidson, 2001).

Thus, for the ideal type professional logic, we identified the ways in which work would be organized if it was controlled only by the profession (Freidson, 2001). In this ideal type, professionals rely on abstract knowledge to conduct their practice solely or in partnership with others of the same profession, allowing for professional control over the content and organization of work. Physicians in this ideal type, for example, would practice medicine alone or in association with other physicians; they would not be employees. Credentialing and educational programs would be controlled by the profession as well as the determination of quality services and products and their pricing. If we consider the ideal professional type in accounting, this means that the accounting profession would solely determine appropriate accounting standards; there would not be government regulation other than enforcing professionally developed standards. This conceptualization recognizes that, as Larson (1977) discusses, professional practice requires state enforcement of occupational privilege and, consequently, the state arbitrates interprofessional competition. In this view, occupational jurisdictions are claimed by professional groups but how and to what extent the state intervenes determines the resolution of such competition (Abbott, 1988).

In the ideal type corporate logic, work is encoded in routines of the firm and is subject to the administrative control of managers in a hierarchy (Thornton, 2004). Professionals are employees of the firm and managers, not professionals, determine the content and organization of work as well as the necessary credentials and education. Corporate managers, who may be non-professionals, have responsibility for evaluating the performance of all workers (including professionals). Managers also determine the appropriate quality of services and products offered as well as their price. In this ideal type logic, dentists, for example, would be employees of a corporation with

managers determining which employees do what, the training they need, and the procedures to be followed. Managers could decide to substitute a lower paid worker to do dental work if they felt corporate safety and quality standards could still be met.

In the ideal state logic, the government takes direct responsibility for professional work as opposed to ratifying professional desires. Professionals are employees of the state and thus the state controls professional knowledge, the credentials for professional practice, and the organization of work. Government managers determine the appropriate quality standards as well as the price for services and products. Physicians in this ideal type would be state employees whose requisite credentials and training are determined by the state. The division of medical tasks among various state employees would be determined by government managers who also have the responsibility for ensuring the quality of medical care.

Finally, in the ideal market logic, there is free and unregulated competition with consumer preferences and choice determining success (Freidson, 2001). Professional knowledge is widely available and there are no specific credentials or educational requirements for professional work. In an important sense, the market logic reflects the absence of controls by the professions, corporations, or the state. The practice of law, in this ideal type, would be open to everyone. The purchasing public alone would determine the appropriate preparation for lawyers based on choices they make in retaining a lawyer. The price of legal services in this ideal type would be determined between lawyers and clients themselves within the context of a broader market in which there are a large number of available lawyers and potential clients.

Applying these four logics to the current literature implies that different logics could organize professionals in different situations such as various geographical locations or types of organizations. Alternatively, the work of different professionals in the same geographical location or organization could reflect different logics. In all cases, multiple logics coexist because the influence of different logics is divided up in some way. We label this process of dividing up as "segmenting," and suggest that it is fundamental to understanding how multiple logics coexist. Multiple logics thus may coexist by segmenting their impact on different actors, geographical communities, or types of organizations.

Extending the concept of segmenting to the level of professional work implies that professional work may be segmented between different logics. Theoretically, this is possible if some parts of professional work reflect attributes from one logic while other parts reflect attributes from different logics. While most of the literature has assumed that logics are holistic, there has been some recognition that logics are comprised of decomposable component parts

that can be recombined in different ways (e.g., Thornton & Ocasio, 2008). This suggests that the work of a single group of professionals may reflect the influence of attributes from different logics.

In the next section, we apply this framework to a historical case study of U.S. pharmacists. Through our empirical study, we develop a new conceptualization of professionals and professional work as guided by a constellation of institutional logics derived from broader society. We set out this conceptualization as a theoretical foundation to explicate the relationships among coexisting logics, thus allowing for the simultaneous influence of multiple logics on professionals and their work.

We note that the terms pharmacy, pharmacists, and pharmacies may be confusing. We use “pharmacy” as a general term referring to the “area in the health sciences that deals with the preparation, dispensing, and appropriate use of medicines” (U.S. National Institute of Health, 2008), “pharmacists” as the people who engage in the practice of pharmacy, and “pharmacies” as the stores where medications are sold to customers.

Method

We were drawn to the history of pharmacy because of our interest in investigating a setting where professional practice appeared to be influenced by multiple institutional logics. Our initial observations suggested that pharmacists were consistently faced with pressures to meet professional standards, sell sufficient product to sustain a business, work within large bureaucratic organizations, and also adhere to strict government regulations. With further investigation into the history of pharmacy, we learned about a profession where running a business had consistently been combined with professional duty and adherence to state imposed regulations. Our preliminary research thus confirmed that this was a setting where multiple logics coexisted and concurrently affected professional behavior.

Our research goal was to understand how pharmacists were influenced by these multiple logics over time. Building on the conceptualization of institutional logics as embodied in practices (Friedland & Alford, 1991; Thornton & Ocasio, 2008), we began to develop a database for our research by systematically examining historical records that described pharmacists’ practices. We defined practices as “usual, customary, or constant actions or performance” (Oxford American Dictionary, 2005). We focused our attention on the practices of pharmacists working in retail stores since it is in this sector where the majority of pharmacists have traditionally worked (as opposed to hospitals or other health care organizations).

We conducted a library search to locate all available historical accounts regarding pharmacists in the United States. Our starting point was the classic pharmacy history book (Kremers, Urdang & Sonnendecker, 1963), first published in 1940 and based on documents collected by Edward Kremer (King, 1964). In addition to this history text, which was twice subsequently revised and updated, we located all other historical accounts identified through our library search (e.g., Gosselin & Robbins, 1999; Griffenhagen, 2002; Posey, 2003; Williams, 1998). Then, based on the general historical accounts, we identified and reviewed relevant government reports and documents published by the professional association. To broaden our view of pharmacists' work beyond that of professional leaders, we also collected reprinted firsthand historical accounts from a variety of pharmacy periodicals. And finally, we identified and reviewed published surveys on pharmacists' work, U.S. government documents on employment status and economic indicators, and national and state legislation and regulations relevant to the practices of pharmacists.

To gain a more fine-grained and insider view of the profession, we also systematically analyzed editorials (1917-2004) of the *Journal of the American Pharmacists Association* (APhA). In many cases, editorials addressed particular events that had occurred, and we were able to organize our database around these events, thus connecting primary source editorials with secondary historical accounts. In this way, we developed a systematic record of key events during the history of U.S. pharmacists since their national association was founded in 1852. We also recorded the various views of pharmacists, government representatives, or other actors associated with each event. For example, when the government developed an agency to ensure safety standards for medications (Hilts, 2003; U.S. Food and Drug Administration, 2007), we were able to search for and record not only the event of creating the agency but also reactions from pharmacists, the professional association, government, physicians, and the public. In total, we reviewed more than 2,500 pages of text in addition to the historical accounts (see appendix for a list of our data sources).

With our dataset encompassing more than 150 years, we needed a way to organize our data presentation. Therefore, we temporally grouped our data (Langley, 1999) using the five historical eras commonly used in pharmacy history and textbooks. These eras are (a) *Organizing Pharmacy* (1852-1910)—marked by the founding of the national professional association; (b) *Beginning Academic Reform* (1910-1945)—characterized by increasingly stringent educational requirements, which were fueled by the highly publicized Flexner (1910) report that severely criticized pharmacists as a nonprofession; (c) *Dispensing Only Practice* (1945-1965)—marked by the expansion

of the pharmaceutical industry after World War II which transformed pharmacists' work from compounding and preparing medications to measuring and counting already prepared elixirs or tablets; (d) *Clinical Pharmacy* (1965-1990)—marked by the APhA unveiling the concept of a “pharmaceutical center” at its 1965 annual meeting, beginning a shift toward pharmacists as knowledgeable professionals providing advice as well as dispensing medications; and (e) *Pharmaceutical Care* (1990-present)—marked by the advent of the concept of “pharmaceutical care,” which highlights pharmacists' knowledge about medication within a patient-centered and team approach to health care (Hepler & Strand, 1990).

Since our research interest was in understanding the role of multiple logics on professional practice over time, we drew on both the professional and institutional literature to develop an appropriate methodology. As explained in more detail above, we relied on the concept of ideal types as an analytical approach for studying the role of institutional logics (e.g., Freidson, 2001; Light, 1989; Rao et al., 2003; Thornton, 2004; Thornton & Ocasio, 2008). The use of ideal types for data analysis is based on Weber's depiction of systems of legitimate social control; typologies can be created and empirical data evaluated in relationship to the ideal type (Thornton, 2004). As “formal analytical models by which to compare empirical observations across institutions” (Thornton & Ocasio, 2008, p. 119), ideal types facilitate the systematic comparison of empirical variation (Freidson, 2001).

Therefore, the methodological approach we adopted is based on comparing the work of pharmacists as portrayed in our historical dataset with the ideal type for each institutional logic developed in our preanalysis theoretical framework: the professional, state, corporate, and market logics. We employed a modified content analysis of our data (Krippendorff, 2004) by using an analytic narrative (Bates, Greif, Levi, Rosenthal, & Weingast, 1988; Ingram & Rao, 2004). Our goal was to evaluate the extent to which each of the four logics was observable in the usual practices of pharmacists at different points of time.

To do this, we first specified identifiable characteristics of professional behavior that would allow us to describe observable practices in relationship to each of the four logics (e.g., Freidson, 2001; Light, 1989; Rao et al., 2003; Thornton & Ocasio, 1999; Thornton, 2004). The first column of Table 1 shows the nine characteristics we identified which could be considered in parallel across all the logics. The first characteristic, *Control over knowledge* (A) refers to whether task specific knowledge is restricted to a particular group or organization. The second characteristic, *Scope of practice* (B), refers to the particular tasks performed by a professional and whether mechanisms

Table 1. Ideal Type of Institutional Logic and Associated Characteristics of Pharmacy Work

Characteristics	Professional logic	Corporate logic	Market logic	State logic
A. Control over knowledge	Pharmacists rely on abstract knowledge to accomplish their work, restricting knowledge about drugs to the profession.	Knowledge about drugs is proprietary property of organizations.	Knowledge about drugs is widely available to consumers.	Knowledge about drugs is confined to the state.
B. Scope of practice	Tasks pharmacists perform reflect desires of professional association on content and boundaries of work.	Tasks that pharmacists perform reflect management decisions on content and boundaries of work.	Tasks that pharmacists perform reflect customer preferences on content and boundaries of work.	Task that pharmacists perform reflect state-determined parameters on content and boundaries of work.
C. Entry to practice	Credentials determined by professional association.	Credentials determined by management.	No credentials required.	Credentials determined by state.
D. Education/training	Educational programs are controlled by the profession.	Educational requirements or training determined by management.	No specific education or training.	Educational requirements or training determined by the state.
E. Control of work processes	Pharmacists control own work processes according to profession determined standards.	Tasks that pharmacists perform are subject to managerial rules, procedures, and routines.	Pharmacists determine how their work will be done according to market pressures.	Tasks that pharmacists perform are subject to state rules, procedures, and routines.
F. Employment status	Pharmacists are self-employed or in partnership with other pharmacists.	Pharmacists are hired by organizations.	Pharmacists are self-employed.	Pharmacists are hired by the state.
G. Performance standards	Pharmacists determine whether work meets performance standards.	The work of pharmacists is evaluated by corporate managers who may be nonpharmacists.	Consumers evaluate work of pharmacists—market determines success.	The work of pharmacists is evaluated by state officials who may be nonpharmacists.
H. Control of drug quality	Profession has responsibility for safety and efficacy of drugs.	Corporation has responsibility for safety and efficacy of drugs.	There are no external standards determining safety or efficacy of drugs.	State has responsibility for safety and efficacy of drugs.
I. Determination of price	Profession determines price.	Corporation determines price.	Large number of buyers/sellers determines price.	State determines price.

exist to prevent others from doing that work. *Entry to practice* (C), the third characteristic, concerns the way in which a determination is made regarding the necessary credentials for professional practice. The next characteristic, *Education/training* (D), focuses on whether a particular organization controls the educational preparation for a professional group, such as pharmacists. Fifth, *Control of work processes* (E) is concerned with the determination of how work is done. For example, the specific content of tasks or the order in which they are completed could be set by professional standards or by organizational requirements. *Employment status* (F), the sixth characteristic, indicates whether pharmacists are self-employed versus corporate or state employees. This is an important consideration because individuals who report to a superior tend to hold less personal control over work than those who are self-employed. Next, *Performance standards* (G) describes the way in which quality of work is evaluated. The following characteristic, *Control of drug quality* (H) concerns who holds responsibility for the safety and efficacy of products sold by pharmacists. Finally, *Determination of price* (I) refers to whether specific mechanisms exist to control the price of drugs or whether free market principles apply.

Next, we described the work of pharmacists for each of the nine characteristics for each of the four ideal types for the institutional logics (professional, corporate, market, and state). (See Table 1, columns 2-5). To illustrate, the characteristics of work associated with the ideal type for *Control over knowledge* differ by logic. In the ideal type professional logic, knowledge about pharmacy practice (e.g., content, preparation, and administration of medications) is held by the profession. In the ideal type corporate logic, this knowledge is the proprietary property of the organization. In the ideal type market logic, knowledge about drugs is widely available to consumers. And finally, in the ideal type state logic, the knowledge is controlled by the state. Similarly, we considered different ways that work could be organized in relationship to the second characteristic, *Scope of practice*. In the ideal type professional logic, the content and boundaries of pharmacists' work is determined by the professional association. In contrast, the content and boundaries of work are decided by management in the ideal type corporate logic and are established through customer preferences in the ideal type market logic. Finally, pharmacists' scope of practice is determined by the state in the ideal type state logic.

After identifying the ideal type for each characteristic, we turned to our data set to identify and evaluate the usual practices relevant to each of the nine characteristics or attributes (Thornton & Ocasio, 2008). Table 2 shows the sources of data used. Table 3 shows our complete set of numerical evaluations of each attribute for each logic and each of the five historical eras.

Table 2. Sources of Data by Key Characteristics of Pharmacy Work

A. Control of knowledge	Historical accounts, professional association publications, published surveys of pharmacists' practices, and industry reports providing descriptions of tasks that constituted pharmacy work, the impact of the Internet, and professional and legal restrictions in effect during each era.
B. Scope of practice	Historical accounts, professional journal editorials, and professional association publications, which describe and document the impact of the professional association on the legal exclusivity of pharmacists' tasks and ancillary staff tasks and supervision.
C. Entry to practice	Historical accounts and professional journal editorials describing the required standards for using the title of pharmacist.
D. Education/training	Historical accounts and professional journal editorials describing the impact of professional association efforts to legally require specific educational standards for using the title of pharmacist.
E. Control of work processes	Historical accounts, professional association publications, and government labor force surveys indicating the development of professional standards, and the percentage of pharmacists working for large organizations—that is, chain stores, hospitals, home health care, and nursing homes compared to independently owned drug stores.
F. Employment status	Historical accounts and government labor force surveys indicating the percentage of pharmacists who own their pharmacy compared to those that are employees.
G. Performance standards	Historical accounts and professional journal editorials indicating how pharmacists were evaluated including the role of the market in pharmacists' practices and the prevalence of MBA model of general management compared to prevalence of owner-operated drug stores or pharmacists as store managers.
H. Control of drug quality	Historical accounts, professional association editorials, and government documents describing professional efforts and federal and state legislation regulating pharmaceuticals.
I. Determination of price	Historical accounts, professional association editorials, and government-prepared reports on the pharmacy industry describing the pharmaceutical market including the presence of predetermined price schedules, the percentage of retailers organized into chain stores compared to independently owned stores, and the percentage of prescriptions paid for by third parties compared to individual purchasers.

Table 3. Evaluation of Strength of Influence of Attributes by Institutional Logic and Historical Era

	Professional logic	Corporate logic	Market logic	State logic
Era 1 (1852-1910)				
A. Control over knowledge	3	1	3	1
B. Scope of practice	2	1	4	1
C. Entry to practice	3	1	3	2
D. Education/training	1	1	3	1
E. Control of work processes	1	1	5	1
F. Employment status	5	1	5	1
G. Performance standards	2	1	4	1
H. Control over drug quality	2	1	5	2
I. Determination of price	1	1	5	1
Era 2 (1910-1945)				
A. Control over knowledge	3	1	2	1
B. Scope of practice	2	1	3	1
C. Entry to practice	3	1	1	3
D. Education/training	3	1	2	1
E. Control of work processes	2	1	4	1
F. Employment status	5	1	5	1
G. Performance standards	3	1	3	1
H. Control over drug quality	1	1	4	3
I. Determination of price	3	1	3	1
Era 3 (1945-1965)				
A. Control over knowledge	2	2	1	1
B. Scope of practice	3	2	2	2
C. Entry to practice	5	1	1	1
D. Education/training	5	1	1	1
E. Control of work processes	3	2	3	1
F. Employment status	3	2	3	1
G. Performance standards	4	1	3	1
H. Control over drug quality	1	1	2	4
I. Determination of price	3	2	3	1
Era 4 (1965-1990)				
A. Control over knowledge	2	1	2	1
B. Scope of practice	4	3	1	2
C. Entry to practice	5	1	1	1
D. Education/training	5	1	1	1
E. Control of work processes	3	3	2	1
F. Employment status	2	4	2	1

(continued)

Table 3. (continued)

	Professional logic	Corporate logic	Market logic	State logic
G. Performance standards	3	3	3	1
H. Control over drug quality	1	1	1	5
I. Determination of price	1	3	3	1
Era 5 (1990-present)				
A. Control over knowledge	3	1	4	1
B. Scope of practice	4	3	1	2
C. Entry to practice	5	1	1	1
D. Education/training	5	1	1	1
E. Control of work	2	4	2	1
F. Employment status	1	5	1	1
G. Performance standards	3	4	3	1
H. Control over drug quality	1	1	2	4
I. Determination of price	1	3	2	2

Note: 1 = *very little influence*; 2 = *little influence*; 3 = *moderate influence*; 4 = *strong influence*; 5 = *very strong influence*.

Below, we explain our evaluation process in detail and provide examples from three different eras that illustrate the way we developed numerical evaluations.

We measured the strength of each logic by evaluating the extent to which the pharmacists' practices as indicated in historical accounts were consistent with the practices implied by the ideal type of the logic. To do this, we compared the practices empirically present in each era with the practices implied by the ideal type for each attribute of each of the four logics as displayed in Table 1. The closeness of the usual work practices to the ideal type for each logic reflects the strength of each logic in guiding pharmacists' work during each era. We evaluated closeness to ideal type using a standard scale of 1 to 5, where 1 indicates *very low consistency* (or similarity) and 5 indicates *very high consistency* (or similarity).

In the first era, for example, pharmacists sold medications to customers who wanted to buy. Quality was determined by the purchaser. We saw this practice as related to the attribute, *control of drug quality*. In our evaluation, we compared the empirical practice of selling medications to customers with few quality restrictions to the description of ideal type for each logic. In this way, we evaluated the usual practice of pharmacists during the first era as very similar to the ideal type market logic (*market* = 5); very low consistency

with the ideal type professional logic (*professional* = 1); very low consistency with the ideal type corporate logic (*corporate* = 1); and very low consistency with the state logic (*state* = 1). Also in Era 1, historical accounts and employment statistics showed that almost all pharmacists worked as independent business owners in their own small independent drugstores. We connected this practice with the attribute, *employment status*, and evaluated it as very consistent with the ideal type for both the professional and market logics (*professional* = 5; *market* = 5). We evaluated this practice as very low consistency with the ideal type for both the corporate and state logics (*corporate* = 1; *state* = 1).

In the third era, chain drug stores had an increasing presence. In 1947, 8.8% of drug stores were organized into chains (Kremer et al., 1963). We saw this practice as related to the attribute *determination of price* because the consolidation of individual sellers influences price in a market away from supply and demand in favor of the corporation. We also connected the attribute *determination of price* with the professional association's practice of publishing price schedules (beginning in 1924). Although our historical sources indicated that pharmacists did not necessarily adhere to published price schedules, this practice was successfully challenged in 1962 by the U.S. Justice Department (*Northern California Pharmaceutical Association v. United States*, 1962). In our evaluation, we compared both the challenged practice of the profession publishing price schedules, the lack of complete adherence to these schedules, and the extent to which chain drug stores dominated the market during this era to the ideal types for the logics and judged it to be moderately similar to the ideal professional logic (*profession* = 3), somewhat consistent with the ideal type for the corporate logic (*corporate* = 2), moderately similar to the ideal market logic (*market* = 3), and having very low consistency with the ideal state logic (*state* = 1).

Finally, in Era 5, pharmacists expanded their responsibilities to include patient management responsibilities, conducting drug use reviews, and counseling patients but continued to spend the largest portion of their time processing prescription orders. We saw these practices related to the attribute, *control over knowledge*. We evaluated the extent to which these practices collectively reflected pharmacists relying on abstract knowledge to accomplish their work, restricting knowledge to the profession, as moderate (*profession* = 3). We also observed that in 1997 the FDA relaxed direct to consumer drug advertising guidelines resulting in advertising dollars tripling from 791 million in 1996 to 2.5 billion in 2001 (Public Citizen's Congress Watch, 2002). At the same time, information about drugs became widely available through the Internet. We also connected these practices to the attribute, *control of knowledge*, since they both affect the extent to which knowledge is widely

Table 4. Average Strength of Influence of Institutional Logics by Historical Era

	Professional logic	Corporate logic	Market logic	State logic
Era 1 (1852-1910)	2.22	1.00	4.11	1.22
Era 2 (1910-1945)	2.77	1.00	3.00	1.44
Era 3 (1945-1965)	3.22	1.55	2.11	1.44
Era 4 (1965-1990)	2.88	2.22	1.77	1.55
Era 5 (1990-present)	2.77	2.55	1.88	1.55

Note: *low* = 1.00-2.25; *moderate* = 2.26-3.74; *strong* = 3.75-5.00.

available to consumers as expected by the market logic. We evaluated the closeness to the market logic as similar (*market* = 4). We evaluated both the corporate and state logics as low consistency for the attribute *control over knowledge*, since knowledge about drugs was neither the proprietary property of the retail establishment nor the state as would be expected by these logics.

Table 3 provides a full depiction of our evaluations for the five eras. In each case, the two authors conducted the evaluation independently. We calculated our initial interrater reliability at 90%. Our differences were small in each case (one point differences on the 5-point scale) and we resolved them through conversation and debate until agreement was reached. Finally, to provide a single evaluation for each logic, we calculated the mean value for attributes within a logic for each era. These evaluations are indicated in Table 4.

In the next section, we describe and analyze pharmacists' practices in each era. To do this, we develop an analytic narrative for each era and explain the nature of pharmacists' work in relationship to the ideal types for each of the four institutional logics—professional, corporate, market, and state. We also explain our evaluation of all logics collectively, showing the pattern of logics that characterizes each era. To enhance the readability of our narrative, we follow Leblebici, Salancik, Copay, and King (1991) in listing our historical data sources in an appendix.

Pharmacists' History and Analysis

Era 1: Organizing Pharmacists (1852 to 1910)

A large portion of those in whose hands the practice of pharmacy (in the United States) now exists, are not properly qualified for the responsible offices it involves, chiefly by reason of the many difficulties that

impede the acquirement of a correct knowledge of their business. (Preamble to APhA's original constitution, quoted in Kremer et al., 1963, p.190)

This era starts with the founding of the oldest and largest national association for pharmacists, the American Pharmaceutical Association (APhA) in Philadelphia in 1852. The primary reason for its creation was to address serious societal problems arising from the lack of drug standards. The quotation above shows that one of the key mechanisms for improvement was to develop much better education and training for pharmacists. Prior to 1852, pharmacists were primarily merchants who sold medications and other products and there were few regulations with respect to practice.

In this era, the APhA began to take actions that were reflected in stronger influence of the professional logic on pharmacists' practices although the overall level of influence remained low (2.22: see Tables 3 and 4 for evaluations). The association developed a model law designed to impose professional standards and regulate practice. By 1903, all but two states at least partially complied by developing laws restricting who could call themselves a pharmacist. However, dispensing drugs was not limited to pharmacists; a 1904 survey found that physicians often dispensed their own medications. To acquire the recognized professional title, pharmacists were required to apprentice and pass an exam demonstrating their knowledge of drugs but formal education was not required. Most pharmacists gained sufficient knowledge to pass their licensing exams through correspondence schools. In 1900, only 12% of pharmacists were educated beyond an apprenticeship (Kremers et al., 1963). The work of pharmacists mostly involved compounding medications. That meant maintaining a stock of chemicals in various forms and carefully mixing up elixirs or powders based on prescriptions from physicians or their own diagnoses. This work required a significant knowledge base although most acquired this knowledge through experience rather than formal education.

There was no influence from the corporate logic (1.0). Pharmacists of this era owned and operated their own drug stores. They functioned as independent shopkeepers, selling not only medications but also household goods, candy, and various other items to individual customers. Almost all set their own workplace conditions and standards, in small self-managed and nonbureaucratic organizations. They hired clerks to assist in the store, providing training and apprenticeship opportunities. In many cases, stores were sold to former clerks who took over as the new pharmacist.

The strong influence of the market logic (4.11) was reflected in pharmacists' practices. Pharmacists sold their goods and services in a market that was characterized by unrestricted price competition. There were few market controls and a relatively large number of buyers and sellers. In an effort to reduce costs, a small cooperative movement among owners of independent drug stores developed, providing a buying group for those who belonged but more competition for other pharmacists. However, ownership of multiple stores remained relatively rare in this era. In 1900, there were no more than 25 drug stores organized into chains (Williams, 1998). At the end of this era, following the failure of professional efforts to set drug standards, the federal government created the Food and Drug Administration (FDA) to protect public safety and, as a result, new legislation was implemented that placed controls on the labeling of drugs sold by pharmacists ($state = 1.22$).

Summary of Era 1. Our overall evaluation for this era indicates that pharmacists' practices were strongly influenced (4.11) by the market logic, with low influence (2.33) by the professional logic. Our data showed almost no influence from the corporate (1.0) or state (1.22) logics. This pattern reflects our observation that almost all pharmacists were self-employed in small drug stores where they were not affected by managerial structures and systems. Instead, pharmacists were primarily shopkeepers who sold products to those who would buy. They compounded medications for individual customers with few professional or regulatory controls and an overall attitude of "buyer beware."

Era 2: Beginning Academic Reform 1910-1945

(The) American Pharmaceutical Association has given major consideration in its work to pharmaceutical education: to the enactment of adequate legislation to regulate the licensing of pharmacists and the practice of pharmacy . . . The Association has always realized that the commercial ability of the pharmacist is second in importance only to his pharmaceutical training. (*Journal of the American Pharmaceutical Association* [JAPhA], 1928, p. 5)

This quotation shows a primary mission of the APPhA during this era was to increase and enforce standards for the education of pharmacists even as the commercial nature of pharmacy was recognized. Our evaluation of the data indicates that the professional logic strengthened (2.77) during this era due to increased educational standards resulting from legislative initiatives driven by the professional association. A key contributing factor was a 1915 invited

address by Abraham Flexner to the National Conference of Charities and Correction where he explicated the concept of profession that had grounded his highly influential Carnegie Foundation commissioned study on medical education reform (1910).¹ Using his definitional template, Flexner publicly rejected pharmacy as a profession. Further chastising pharmacists, the U.S. government refused to recognize them as professionals in World War I because of inadequate educational standards. In response, the professional association developed new requirements for a 3-year diploma program that was increased to a 4-year degree by 1932. However, states did not automatically develop legislation to enforce these standards, and as a result, many pharmacists continued to enter practice through apprenticeship programs (*state* = 1.44). In spite of efforts by the APhA to restrict dispensing of medications to pharmacists, more than one third of medications were dispensed by physicians. In the 1920s and 1930s, "The work of the day was the compounding and dispensing of medication forms such as table triturates, capsules, powders, some pills, some cachets, ointments, suppositories, and the like" (Archambault, 1982, p. 226). However, by 1940, the proliferation of pharmaceutical manufacturers resulted in only 26% of medications requiring compounding by the retail pharmacist (Kremer et al., 1963). This meant that by the end of this era, pharmacists spent most of their time putting already prepared pills into containers and selling them to customers.

The corporate logic remained insignificant (1.0) in influence. Pharmacists continued to work primarily in small drug stores as the owner and operator. They managed a small staff but were not managed by others. Only a very small number of pharmacists worked in chain drug stores. In 1920, chains comprised 3% of all pharmacies, growing to 6% by 1930, according to trade surveys cited by Williams (1998).

In this era, we evaluated the influence of the market logic as moderate (3.00) as conditions moved away from the free market standard. The federal government increasingly took actions to control the market based on concerns for public safety. As a result, pharmacists were restricted to selling drugs conforming to federal safety standards, in addition to being properly labeled. In 1924, the APhA introduced the first of several price schedules—guaranteeing pharmacists a set profit on the sale of pharmaceuticals and protecting them from supply and demand mechanisms. Also, in response to concerns that chain stores would eliminate small independent pharmacies, a few states restricted the ownership of pharmacies to pharmacists. These attempts to control the market were successfully challenged in 1928 when the U.S. Supreme Court ruled that anyone could own a pharmacy.

Summary of Era 2. Pharmacists' practices reflected moderate influence by the professional (2.77) and market logics (3.00) with no influence by the corporate logic (1.0) and low influence by the state logic (1.44). This pattern can be explained by the increasing emphasis by the professional association on determining educational requirements for pharmacists and by increasing government intervention in the market for medications, primarily to ensure standards of public safety. Since very few pharmacists worked in locations where they reported to a supervisor, the practices of pharmacists continued to be inconsistent with attributes of the corporate logic.

Era 3: Dispensing Only Practice (1945 to 1965)

Once characterized by galencials, empiricism, and the trade school type philosophy, we have advanced to the era of specific and standardized medication, developed on the basis of scientifically designed experiments. The era of the spatula and the mortar and pestle has yielded to an age of electronically controlled machinery mass producing drugs on a scale never dreamt of just a few decades ago. (*Journal of the American Pharmaceutical Association*, 1958, p. 411)

This quotation describes the advances in technology that dramatically changed pharmacists' work from compounding medications to dispensing mass produced drugs. While presented as an "advance" for pharmacy, pharmacists' practices became increasingly routinized during this era, requiring less abstract knowledge. Historians refer negatively to this time as "count and pour pharmacy" (Whitney & Nahata, 2009). The 1952 APhA Code of Ethics continued prohibitions on pharmacists sharing information about medications with patients. Federal legislation was implemented to restrict pharmacists by requiring physician approval for some previously discretionary practices (*state* = 1.44). In most states, the practices of pharmacists were further restricted by antisubstitution laws, requiring pharmacists to provide the exact trade-name drug prescribed by the physician, a reduction in scope of practice. Overall, the influence of the professional logic was moderate (3.11) because these changes were offset by the professional association taking firm control over educational programs, giving pharmacy schools the exclusive right to educate pharmacists. In 1960, the APhA required a 5-year university degree for all new pharmacists, even though this standard was opposed by the chain drug stores because it would increase their costs.

Although the number of pharmacists working in retail drug store chains increased, the overall strength of the corporate logic was low (1.55). The

growing trend toward suburban living meant that the “. . . owners of many long-established pharmacies with fine reputations (are) finding it absolutely necessary to relocate or close their doors. . . . A third of the pharmacies in some metropolitan areas are (were) reported to have gone out of business or moved elsewhere” (*Journal of the American Pharmaceutical Association*, 1958, p. 202). However, the managerial conditions in which most pharmacists worked did not change substantially during this era. Between 1948 and 1960, the number of pharmacists owning a store alone or with a partner remained steady at 52% while only 1.0 to 1.5% of pharmacists worked in settings employing five or more pharmacists (Kremer et al., 1963). Stores did grow larger during this time, as indicated by a drop from 50% to 40% of pharmacies operated by a single pharmacist (Kremer et al., 1963), but pharmacists’ practices were still only minimally influenced by managerialism.

Pharmacists’ practices reflected influence by the market logic to a lesser extent (2.22). There was movement away from a free market ideal as the federal government became increasingly involved in guiding the market for pharmaceuticals (*state* = 1.44). In 1952, drug products were categorized into “prescription” and “over the counter” (OTC), eliminating pharmacists’ ability to dispense certain classes of drugs without a physician’s prescription. In 1962, the government tightened its control over new drugs and imposed requirements of effectiveness (not only safety). In the same year, the U.S. Justice Department won an antitrust complaint against professional associations that had been controlling prices pharmacists could charge for medications. Also, pharmacies were beginning to consolidate as part of the rise of chain stores; in 1950, only 7.7% of retail stores were chains, increasing to 10% by 1965 (Williams, 1998).

Summary of Era 3. Our evaluation of the data shows that pharmacists’ practice reflected moderate influence of the professional (3.22) logic and also lesser influence by the market (2.11), corporate (1.55), and state (1.44) logics. This pattern can be explained by the reduced need for professional knowledge as pharmacists shifted from compounding medications to only dispensing them. However, counteracting this decrease in professionalism, the pharmacists association gained control over entrance and educational standards. At the same time the market logic weakened as fewer pharmacists were self-employed and the state took a stronger role in regulation. And although still minimal, the corporate logic began to be reflected in practices as larger pharmacies in suburban areas became the norm. The state logic’s influence was significant in the control over the quality of drugs.

Era 4: Clinical Pharmacy 1965 to 1990

Why don't we look for ways to complement our professional stature rather than looking for ways to make us even more ordinary merchants? Let's figure out how we can be indispensable health professionals. If we become such professionals, then I can tell you, as surely as night follows day, that we will finally get the recognition and remuneration we deserve for our services. (*Journal of the American Pharmaceutical Association*, 1964, p. 65)

This era was characterized by the ideal of clinical pharmacy with a renewed focus on pharmacists as knowledgeable professionals. Key members of the APhA encouraged an expanded role for pharmacists that allowed them to provide advice to their customers as well as pharmaceuticals. Overall, our data show that during this era pharmacists' practices reflected moderate influence (2.88) of the professional logic. In 1969, the APhA revised its Code of Ethics to eliminate the old prohibition on sharing information with patients. This provided the opportunity for pharmacists to give advice as part of their interactions with customers. Educational standards continued to increase and the curriculum was revised in the 1970s to incorporate this new orientation on the provision of information. The federal government recognized a clinical role for pharmacists in 1974 when it began requiring that pharmacists conduct monthly reviews in nursing homes. As well, increases in professionalism were realized through legislation allowing pharmacists to use their professional judgment and substitute generic drugs for more expensive trade name drugs prescribed by physicians.

The influence of the corporate logic became more apparent (2.22) during this era. Pharmacists continued to transition from being owner practitioners to employees of large organizations. "In less than 20 years, pharmacy (has) changed from a profession characterized by practitioners who were pharmacy owners to one in which pharmacists are predominantly employees." (U.S. Department of Health, Education, and Welfare, 1982, pp. viii-1). In 1956, almost half of pharmacists owned a store alone or with a partner (U.S. Bureau of Labor Statistics, 1957) while only 20% did in 1970 (U.S. Bureau of the Census, 1973), falling to 9.4% by 1983 (U.S. Bureau of the Census, 1981). As employees, pharmacists were increasingly influenced by managerial practices. In the early part of this era, chain drug stores followed a professional model ensuring that pharmacists were managed by more senior pharmacists. In spite of concerns from the APhA, a more managerial approach was implemented during the 1980s and pharmacists began reporting to

“managers and health care administrators who are not pharmacists” (Bezold, 1985, pp. 173-174). It became a managerial decision whether to hire pharmacist technicians and, within the constraints of legislation, how many technicians to hire.

The influence of the market logic became lower (1.77) as the consolidation of buyers and sellers moved the system away from a free market model. In 1970, 12.7% of retail pharmacies were chains, increasing to 39% by 1987 (Williams, 1998). By 1989, almost 40% of all prescriptions were paid for by health insurance plans that could negotiate prices (U.S. Department of Health & Human Services, 2000). Control over drugs continued to be monopolized by the state as the FDA increased requirements for approval of new drugs. As well, pharmacists were influenced by the rise of consumerism. In 1976, the U.S. Supreme Court ruled in favor of a consumer group challenging a state board of pharmacy’s right to restrict price advertising for prescription drugs. Patients began to think of themselves as consumers, challenging the view that responsibility for a person’s health rests primarily with a physician or health professional rather than the patient. “Consumer groups visit pharmacies and determine the hours of operation and a listing of services (for example, whether credit is given, whether patient profiles are maintained, whether a delivery service is available)” (Wertheimer, 1985, p. 106). Pharmacists were forced to meet these customer group standards or their sales would decrease.

Summary of Era 4. Pharmacists’ practices reflected moderate influence by the professional (2.88) and the corporate (2.22) logics but less by the market (1.77) and state logics (1.55). Our rating of the professional logic was moderate because, in addition to controlling educational standards, pharmacists were able to increase the value of their professional knowledge. While still low overall, the influence of the corporate logic strengthened because pharmacists as employees were subjected to increasing levels of management. And the market logic became less influential as buyer and seller consolidation was only partially offset by increases in the availability of information to consumers. The state logic became more influential than previously as the government increased its regulation of pharmaceuticals.

Era 5: Pharmaceutical Care 1990 to Present

One of APhA’s primary goals is to encourage consumers to talk with their pharmacist and other health care providers and work together to promote wellness, prevent disease, and to assess, monitor, initiate, and modify medication use to assure that drug therapy regimens are safe

and effective. (*Journal of the American Pharmaceutical Association*, 1998, p. 117)

This quotation shows continued attention to consumers and market forces even as pharmacists began to focus on an advanced role as health professionals bringing expert knowledge about medicines (prescription, behind-the-counter, OTC preparations) to a collaborative health care team. In this era, the overall influence of the professional logic was moderate (2.77). The APhA developed and spread the concept of pharmaceutical care, which was defined as “the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient’s quality of life” (Hepler & Strand, 1990, p. 539). Pharmaceutical care represents a rhetorical strategy on the part of the pharmacy profession to construct itself in a way that recognizes the importance of pharmacists’ abstract knowledge within a patient-centered team approach to health care. This emphasis on pharmacists as knowledgeable professionals has been accepted and legitimated by both state and federal governments through requirements and funding for patient counseling, and by insurance companies through coverage for counseling services from pharmacists. The professional association increased educational standards in this era with the result that pharmacists required a 6-year university degree for entry to practice. However, in spite of these efforts pharmacists continued to spend 48% of their time processing prescription orders (Schommer et al., 2006).

The influence of the corporate logic as reflected in practice increased to moderate (2.55). Almost all pharmacists were employees. Survey data from 1998 indicates that only 3% of pharmacists reported being self-employed (U.S. Bureau of Labor Statistics, 1998). Moreover, pharmacists increasingly worked for large organizations; 71% of pharmacists worked in chains in 2004 compared to 56% in 1990 (Mott et al., 2006). Managerially controlled division of labor in large pharmacies has resulted in large productivity gains. Although pharmacists held responsibility for dispensing medications and providing appropriate information to patients, they were increasingly assisted by pharmacy technicians and mechanized systems including robots. These efficiencies supported a 44.1% increase in prescription volume in this era (U.S. Department of Health & Human Services, 2000). Overall, we see that within large retail stores, pharmacists’ work was routinized. As employees, they followed organizational rules and procedures and worked within a hierarchy of authority where they were managed by others—often by nonpharmacists.

The influence of the market (1.88) and state (1.55) logic can be observed in pharmacists’ practices. The market for pharmaceuticals was controlled by

government safety standards and by large buying groups such as insurance companies that the federal government relied on to administer the pharmacy benefits added to Medicare. However, the FDA reduced restrictions on drug approval in 1997 and the number of formerly prescription drugs available over the counter (OTC) increased. Consumer choice is often restricted by insurance plans or government assistance programs. And the market is relatively distant from that of a “free market” since chain retailers (including drug stores, supermarkets, and mass merchants) comprised 61% of the stores in 1998 (National Association of Chain Drug Stores, 1999). The two largest chains had 4,000 stores each and dispensed 41% of total prescriptions (U.S. Department of Health & Human Services, 2000). However, no formal price controls existed and the number of OTC drugs continued to escalate with the result that many medications were sold much like any other household item, without a pharmacist. The relaxation of federal prohibitions on pharmaceutical companies’ advertising combined with availability of pharmaceutical information through the Internet resulted in more knowledgeable consumers participating in the market.

Summary of Era 5. Pharmacists’ practices reflected moderate influence by both the professional (2.77) and corporate (2.55) logics as well as somewhat lower influence by the market (1.88) and state (1.55) logics. This pattern can be explained by very strong control over educational standards by the professional association and an increased ability for pharmacists to use their abstract knowledge as part of their normal work. As employees of large companies, pharmacists’ practices were strongly influenced by principles of managerialism. And, at the same time, the influence of the market logic became more apparent as information about medications became widely available through television advertisements and the Internet. The role of the state logic remained largely confined to drug regulation.

Discussion

In our study we saw that the professional work of pharmacists was simultaneously guided by different logics. Our analysis of pharmacists’ practices across eras reflected influence from the professional, corporate, market, and state logics. In the first era, our analysis showed very strong influence of the market logic with only low level influence from the professional logic and virtually no influence from the corporate or state logics. In the second era, we observed significant influence of both the professional and market logics. In the third and fourth eras, the professional logic remained significant while concurrently the market logic decreased and the corporate logic increased in

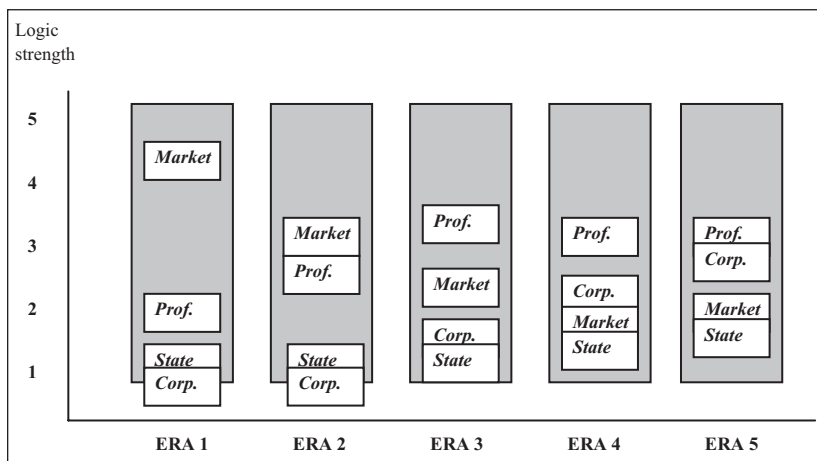


Figure 1. Constellations of institutional logics by historical era

strength. In the fifth era, we found the corporate logic and the professional logic both were moderately influential while the market and state logics were less influential. We illustrate the patterns or constellations of logics we observed for each era in Figures 1 and 2.

We use the term constellation to describe the combination of institutional logics guiding behavior at any one point of time. Constellation means a configuration or position of “stars” (or potentially other elements) in regard to one another (Oxford American Dictionary, 2005). This term is conceptually useful because a constellation is composed of items that, when viewed from a particular perspective, can be identified as a pattern. Viewing logics as constellations guiding behavior contrasts with previous explanations that have conceptualized variation in logics in terms of different geographical locations, different actors, or different types of organizations (e.g., Lounsbury, 2007; Marquis & Lounsbury, 2007; Reay & Hinings, 2009). While others have recognized that society is composed of a variety of “meta-logics” (Scott, 2008b; Thornton & Ocasio, 2008), they have not explicitly theorized about the relationships among the multiple logics and their collective influence on a social actor. Viewing coexisting logics as constellations helps us to see the ways that societal logics can combine to simultaneously influence professional work, since the concept of a constellation captures the pattern in which these multiple logics are combined at a given time. We identified three different patterns: a single dominant logic with three relatively dormant logics

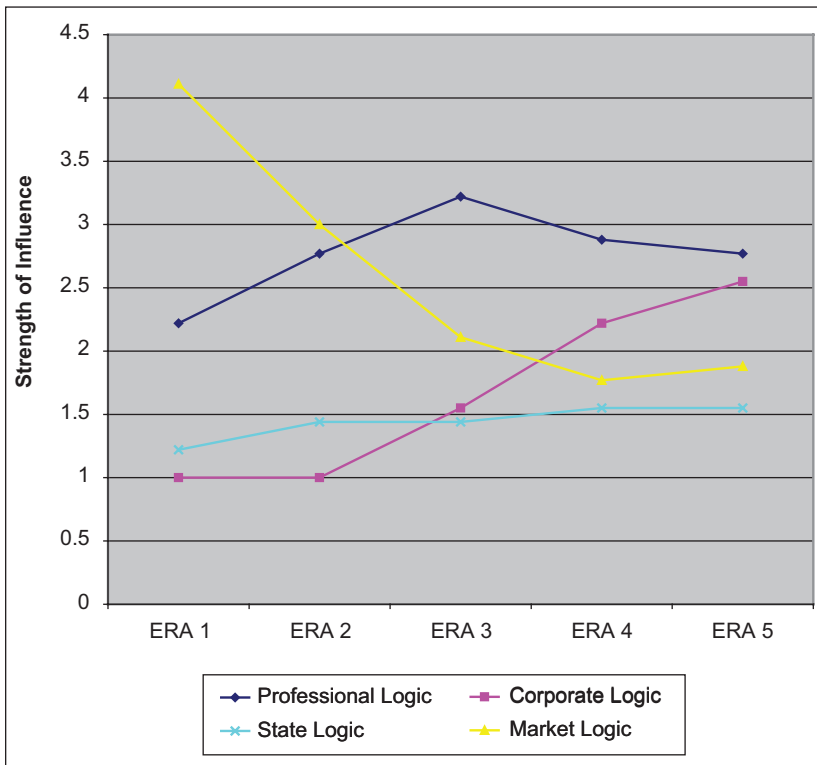


Figure 2. Average strength of influence of institutional logics by historical era

(Era 1), two relatively strong logics with neither one dominant and two weaker logics (Era 2 and Era 5), and a single relatively strong logic and three weaker logics (Eras 3 and 4).

In our study, we identified only one example of a constellation dominated by a single logic. In the first era we observed that pharmacists' practices almost exclusively reflected the market logic with a low level of influence from the professional logic and almost no influence by the corporate and state logics. By using the concept of a constellation of logics, we provide an alternative model for recognizing the potential role of secondary or dormant logics. This model, then, shows a theoretical foundation for studies focusing on secondary logics that challenge and usurp previously dominant logics (e.g., Hensmans, 2003). By conceptualizing a dominant logic as only part of an arrangement of logics in a constellation, the role of secondary logics can be highlighted.

In the second type of constellation we identified, two logics exercise relatively equal and significant influence on behavior. Similar to constellations dominated by a single logic, this type of constellation may also contain other logics with significantly less influence. In Era 2, the work of pharmacists was largely guided by both the professional and market logic with relatively low influence from the corporate and state logics. In Era 5, the work of pharmacists reflected strong influence by the professional and corporate logics. While the existing literature has recognized that multiple logics coexist, the tendency has been to portray the existence of two significant logics as a temporary transitory state between different dominant logics. Rather than resolving to a single dominant logic, we saw in our study of pharmacists a very lengthy period of time where two different logics coexisted and concurrently influenced the practice of one actor.

The third type of constellation we identified had one logic exercising moderate influence and three with less influence. In this constellation type, one of the weaker logics was more important than the others. In Era 3, the professional logic had a moderate influence on practice while the weakening market logic had substantially more influence than the corporate or state logics. In Era 4, the professional logic remained moderate in influence but the substantial influence of the market logic shifted to the strengthening corporate logic. Conceiving of multiple logics as a constellation has the advantage of conceptualizing the impact of logics with less overall significance. The state logic is a striking example of this. While the average influence of the state logic was low in both of these eras and remained low throughout our case study, the responsibility for the safety and efficacy of drugs became a governmental mandate with wide reaching effects for pharmacists and the public.

By conceptualizing logics in terms of constellations, our attention was drawn to understanding the dynamics between or among logics within the constellation. By examining critical events over time, we were able to focus on how work practices reflected the interaction of one logic with others. Through this process, we found evidence of both competitive and cooperative relationships among the logics. Competitive relationships imply that one logic's "win" is another logic's "loss"; that is, increases in the strength of one logic occur with a corresponding decrease in another logic. Competition between logics thus suggests that there must be equal gains and losses from the available total "pie." In contrast, a cooperative relationship between institutional logics implies that a "win-win" for both logics is possible. In this case, increases in one logic do not necessarily mean a corresponding decrease in the strength of another logic.

Even though most of the established literature is based on an assumption of competition between logics, there have been few explanations of how

competition can persist. Instead, the literature has focused on the resolution of conflict when a new logic becomes dominant. Our findings show that, at least in some situations, multiple logics may coexist competitively for lengthy periods of time. We observed that competition between logics persisted because practices were essentially segmented; different practices were guided by different logics. We observed examples of segmenting in every era. For example, by Era 3 educational requirements for pharmacists had been separated from other work conditions and were organized exclusively by the professional logic. This separation meant that the professional association could determine standards for entry to practice independently of other considerations. We saw evidence of the robustness of this segmenting when chain stores objected to increasing educational standards based on arguments of increased costs associated with the corporate logic, but their objections were overruled by the established rights of the professional association. Another example of segmenting is evident in Era 4 when large chain and grocery stores implemented a managerial model of supervision that resulted in pharmacists reporting to nonpharmacists. This practice was antithetical to the professional logic but could not be successfully opposed by the APhA because managerial relationships were under the control of employers. Segmenting was a mechanism that allowed multiple competitive logics to coexist by dividing up the available “pie.”

Although competitive relationships are documented in the established literature (e.g., Greenwood & Suddaby, 2006; Hensmans, 2003; Freidson, 2001), cooperative relationships have received little attention. Our findings indicate the relationship between logics may also be cooperative. Rather than logics competing to divide up the available “pie,” the “pie” itself may be expandable. We saw two different ways relationships between logics may be nonzero sum. First, we observed instances in which the relationship between logics was facilitative. This was clear when different logics simultaneously gained increased influence in guiding pharmacists’ practices. That is, one logic increased in strength during an era without a coinciding decrease in other logic(s). For example, in Era 5 we observed the increasing influence of the market logic dealing with control of knowledge supported the professional logic. Customer knowledge of medications brought about by widespread access to the Internet and media advertisement of pharmaceuticals facilitated increases in the abstract nature of pharmacists’ work, as pharmacists interacted with better educated consumers interested in pharmacist counseling. Consequently, increased customer access to information supported pharmacists’ efforts to control their work through increasing the value of abstract knowledge.

Second, we observed that nonzero sum relationships between logics can also be additive. In these situations, work practices were influenced by multiple logics. For example, in Eras 2 and 3 pharmacists had to meet both professional and market performance standards. To maintain his position, a pharmacist had to be both professionally competent as well as meet customer expectations, which might include additional dimensions. Convenience, for example, is an expectation consistent with a market logic but not a professional logic. During Era 3, many professionally competent pharmacists went out of business because their locations were inconvenient for their newly suburban customers. This example illustrates how standards consistent with different logics do not necessarily present competing mandates.

Conclusions

We wanted to understand how the work of a single profession can be collectively influenced by different institutional logics. We developed the concept of a constellation of institutional logics (a set of logics in a recognizable pattern) where it is the combination of logics that influences professional work. In our study, we identified three different types of constellations: (a) a constellation where one logic is dominant over the others, (b) a constellation where two logics exercise relatively equal and significant influence on behavior, and (c) a constellation where one logic exercises moderate influence and others show some, but less influence. The concept of a constellation of logics is important because it helps to explain not only the situation of a dominant logic (e.g., Scott et al., 2000) but also situations where multiple societal level logics coexist and collectively affect professional work for lengthy periods of time. It is through consideration of the second and third types of constellations that we were able to highlight the role of both competitive and cooperative relationships among logics. Overall, we see that to understand professional work, it is important to focus attention not only on apparently dominant logics but also on the full set of relevant institutional logics.

Our study contributes to the literature on professions in three ways. First, we provide a new theoretical framing that helps to understand the development of professionals over time. The conceptualization of constellations provides a way to articulate the different logics that may be reflected in professional work. In the extant literature, professional work is described by decreasing levels of professionalism and corresponding increases in other forces such as consumerism or corporatization. Professionals are portrayed as having lost their privileged economic and social position (e.g., Freidson, 2001; Krause, 1996). The value of the concept of constellations is to remind

us that even though a logic may no longer be dominant, it still may be reflected in the organization of professional work. Leicht and Fennel (2008), for example, discuss how market-based solutions are increasingly applied to professional practices. Their portrayal is consistent with movement from a constellation in which one logic is dominant to one in which work reflects relatively equal and significant influence from two different logics. In calling attention to the arrangement of logics, the concept of constellations highlights the relative influence of logics, promoting attention both to multiple logics with relatively equal influence as well as those with less influence.

Second, our work extends the literature on professional work by articulating the mechanisms by which work can be affected by multiple logics. Conceptualizing institutional logics as existing in constellations provides a theoretical foundation to explicate the relationship among coexisting logics, allowing for their simultaneous influence on professionals and their work. Our research suggests that both competitive and cooperative relationships among multiple logics exist. Logics may be competitive but still be jointly reflected in professional work because of segmenting; some aspects of work may be guided by one logic while others are guided by alternative ones. Consequently, we may observe that some dimensions of professional work have undergone change while others have not (Leicht & Fennel, 2008). For example, educational standards may be set by the professional association, while workplace standards are set by the corporation. Competitive relationships characterized by segmenting allow multiple institutional logics to coexist for a lengthy period of time. In addition to competitive logics coexisting through segmenting, our analysis suggests that cooperative relationships also explain the simultaneous influence of multiple logics on professionals. Nonzero sum relationships between logics imply that the impact of multiple logics may not be to force choices between competing demands but rather to increase the number of demands that must be met. For example, professionals may have to meet standards set by the professional association and also meet corporate regulations for productivity or efficiency.

Finally, we contribute to the literature on professional work by developing abstract characterizations of four logics that may be used in other studies of professional work. In doing so, we have provided focused attention to the ideal type state logic apart from its role in enforcing occupational privilege. While the influence of capitalistic organizations and the market have been recognized in studies of professional development, the possibility of professional work reflecting a state logic has not been highlighted. We know that in some countries the state can intervene in public affairs and take direct responsibility to guarantee both the well-being and health of its citizens (Currie &

Guah, 2007). Sweden, for example, has had a nationalized system of pharmacy under government ownership (with government employed pharmacists), which guaranteed access for its citizens (Lindberg & Adolfson, 2007). Future studies could draw on our conceptualization of logics to investigate similarities and differences across national settings.

Our work also contributes to the institutional literature on logics. First, we articulate that segmenting is the mechanism underlying empirical examples of multiple coexisting logics in the current literature (e.g., Lounsbury, 2007; Reay & Hinings, 2009) and extend this mechanism to the level of the individual actor. Rather than different organizations, geographical communities, or actors being guided by different logics, our research suggests that some dimensions of professional work may reflect one logic while others reflect different ones. Second, we provide a reappraisal of theory that multiple logics will inevitably compete with other logics by explicitly considering that the relationship among multiple logics may have cooperative aspects. We observed evidence of two different types of nonzero sum (cooperative) relationships: facilitative and additive. Facilitative relationships between logics suggest that shifts in work practices consistent with an aspect of one logic may support changes consistent with an alternative logic, such as when the demands of consumers for information supports professional control over abstract knowledge. Additive relationships suggest that cooperation may not necessarily mean finding a common way of working from different logics (Reay & Hinings, 2009). Neither does it rely on identifying practices that can be adopted by those adhering to alternative logics for different reasons (Lounsbury, 2007). Instead, cooperation may mean that the impact of coexisting institutional logics is simply to increase the number of demands that social actors must satisfy.

Since our focus was on understanding professional work in terms of the relationships among multiple institutional logics, we have given less attention theoretically to how new practices evolved. Consequently, we have not explicitly addressed the role of agency in change processes involving multiple logics. Still, our analytic narrative does suggest that social actors attempt to use societal logics as a “tool box” to fit their interests (Swidler, 1986). For example, the newly established APhA took actions to combine attributes of the professional logic with those of the market logic. By establishing state associations to lobby for legislation distinguishing a pharmacist from a merchant, the APhA worked to have the profession rather than the market control who could call themselves a pharmacist. As we saw in the case of pharmacists, actors’ preferences are limited by other factors, including their relative power vis-a-vis other actors. Future research could focus on illuminating the relationship between agency and logics when multiple logics exist and hold

the potential to influence professional work. We suggest that such research is certainly warranted.

In this article, we developed a model of constellations of institutional logics and used this theoretical conceptualization to understand the professional work of pharmacists over time. Compared to other professionals, pharmacists' claims to status have historically been more tenuous because of their long association with business and the market. However, there have been widespread changes in the last 30 years in the organization of professional work, which has resulted in attempts to maintain professionalism against alternative logics (e.g., Kitchener, 2002; Reay & Hinings, 2005). As a result, we suggest our findings are applicable to other professions beyond our case study of pharmacy. Segmenting, for example, can help explain how professionals such as physicians can be both highly professionalized and also influenced by managerialism. Physicians may modify their practice to meet managerial requirements of insurers or Health Maintenance Organizations (HMOs). However, they also follow professional standards and norms, and use these as guiding principles for conducting their medical work. At any one point in time, it is the combination of institutional logics and the relationship between them that provides important information about the nature of professional work.

By conceptualizing institutional logics as part of a constellation characterized by both competitive and cooperative relations, we have provided an alternative model for understanding the coexistence of multiple logics and their influence on professionals and their work. In our approach, we draw attention to practices but in doing so have given little attention to the symbolic aspects of institutional orders that have been linked to change in other contexts (e.g., Davis, Diekmann, & Tinsley, 1994; Davis & Greve, 1997). While we have privileged practices over symbols, we see that this is appropriate in a study of pharmacy that is by nature practice oriented. This may not be the case in other contexts. We encourage researchers to apply our model of logics to other professional settings—particularly in fields that are at a juncture of different societal logics. It is through further research that the generalizability of our conceptualization can be considered. In short, we hope that our model of constellations of institutional logics will provide a new line of sight that encourages further theoretical development about logics and the way in which their influence is reflected in professional work.

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1. The Carnegie Foundation commissioned Flexner report (1910) severely criticized medical education in the United States and Canada. Subsequent to its publication, many medical schools were closed and higher admission and graduation standards were enacted.

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